

Python Programming

Problem Solving Activity – 2

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Control Flow & Functions

Topics Covered: if, if-else, if-elif-else, nested conditions, for, while, break, continue, pass, range(), enumerate(), zip()

Question 1 – Student Result Analyzer

Algorithm / Pseudocode :

1. Define lists names[] and marks[]
2. Initialize valid_count = 0
3. FOR each (index, name, mark) using enumerate() and zip():
 - a. IF mark < 0 OR mark > 100:
PRINT "Invalid marks for - Skipped" CONTINUE to next student
 - b. IF mark is between 80 and 100:
result = "Excellent"
 - ELIF mark is between 60 and 79:
result = "Good"
 - ELIF mark is between 40 and 59:
result = "Average"
 - ELSE:
result = "Fail"
 - c. PRINT index, name, mark, result
 - d. Increment valid_count

4.PRINT "Total Valid Students: valid_count"

Source Code :

```
names = ["Arun", "Bala", "Charan", "Divya", "Esha"]
marks = [85, 72, 38, 105, 91]
valid_count = 0

for i, (name, mark) in enumerate(zip(names, marks), start=1):
    if mark < 0 or mark > 100:
        print(f"Invalid marks for {name} - Skipped")
        continue
    if mark >= 80 and mark <= 100:
        result = "Excellent"
    elif mark >= 60 and mark <= 79:
        result = "Good"
    elif mark >= 40 and mark <= 59:
        result = "Average"
    else:
        result = "Fail"
    print(f"{i}. {name} - {mark} - {result}")
    valid_count += 1
print(f"Total Valid Students: {valid_count}")
```

Output :

```
PS C:\Users\Administrator> cd "C:\Users\Administrator\OneDrive\Documents\Python sem3"
PS C:\Users\Administrator\OneDrive\Documents\Python sem3> python P1A2.py
1. Arun - 85 - Excellent
2. Bala - 72 - Good
3. Charan - 38 - Fail
Invalid marks for Divya - Skipped
5. Esha - 91 - Excellent
Total Valid Students: 4
```

Question 2 – Electricity Bill Calculator

Algorithm :

1. Read n = number of consumers
2. Create empty lists: consumer_names[], consumer_units[], messages[]
3. FOR i in range(1, n+1):
 - a. Read consumer name
 - b. Read units consumed
 - c. IF units == 0: Append "0 units entered. Processing stopped." to messages[] BREAK out of loop
 - d. IF units < 0: Append "Invalid units for - Skipped" to messages[] CONTINUE to next consumer
 - e. ELSE: Append name to consumer_names[] Append units to consumer_units[]
4. FOR each (name, units) in zip(consumer_names, consumer_units):
 - a. Determine rate using if-elif-else based on units
 - b. bill = units * rate
 - c. PRINT name, units, bill
5. FOR each msg in messages[:]: PRINT msg

Source Code :

```
n = int(input("Enter number of consumers: "))
consumer_names = []
consumer_units = []
messages = [] # holds "Invalid" / "stopped" messages, in the order they occurred
for i in range(1, n + 1):
    name = input(f"Consumer {i}: ")
    units = int(input("Units: "))
```

```

if units == 0:
    messages.append("0 units entered. Processing stopped.")
    break
if units < 0:
    messages.append(f"Invalid units for {name} - Skipped")
    continue
consumer_names.append(name)
consumer_units.append(units)
for name, units in zip(consumer_names, consumer_units):
    if units <= 100:
        bill = units * 2
    elif units <= 200:
        bill = units * 3
    elif units <= 300:
        bill = units * 5
    else:
        bill = units * 7
    print(f"{name} - {units} units - Bill: \u20b9{bill}")
for msg in messages:
    print(msg)

```

Input :

```

PS C:\Users\Administrator\One
Enter number of consumers: 4
Consumer 1: Arun
Units: 85
Consumer 2: Bala
Units: 175
Consumer 3: Charan
Units: -20
Consumer 4: Divya
Units: 0

```

Output :

```
Arun - 85 units - Bill: ₹170  
Bala - 175 units - Bill: ₹525  
Invalid units for Charan - Skipped  
0 units entered. Processing stopped.
```

Question 3 – Student Attendance and Performance Monitor

Algorithm / Pseudocode :

1. Define lists names[], attendance[], marks[]
2. Initialize eligible_count = 0
3. FOR each (index, name, att, mark) using enumerate() and zip():
 - a. IF att or mark is outside valid range (0-100):
CONTINUE to next student
 - b. IF att >= 75: (nested condition)
IF mark is between 80 and 100:
grade = "Distinction"
ELIF mark is between 60 and 79:
grade = "First Class"
ELIF mark is between 40 and 59:
grade = "Pass"
ELSE: grade = "Fail"
PRINT index, name, att, mark, grade Increment eligible_count
ELSE: PRINT index, name, att, "Not Eligible"
4. PRINT "Total Eligible Students: eligible_count"

Source Code :

```
names = ["Anu", "Bharath", "Chitra", "Deepak", "Farah"]  
attendance = [92, 68, 85, 45, 78]
```

```

marks = [88, 55, 76, 32, 91]
eligible_count = 0
for i, (name, att, mark) in enumerate(zip(names, attendance, marks), start=1):
    if att < 0 or att > 100 or mark < 0 or mark > 100:
        continue
    if att >= 75:
        if mark >= 80 and mark <= 100:
            grade = "Distinction"
        elif mark >= 60 and mark <= 79:
            grade = "First Class"
        elif mark >= 40 and mark <= 59:
            grade = "Pass"
        else:
            grade = "Fail"
        print(f"{i}. {name} - Attendance: {att}% - Marks: {mark} - {grade}")
        eligible_count += 1
    else:
        print(f"{i}. {name} - Attendance: {att}% - Not Eligible")
print(f"Total Eligible Students: {eligible_count}")

```

Output :

```

1. Anu - Attendance: 92% - Marks: 88 - Distinction
2. Bharath - Attendance: 68% - Not Eligible
3. Chitra - Attendance: 85% - Marks: 76 - First Class
4. Deepak - Attendance: 45% - Not Eligible
5. Farah - Attendance: 78% - Marks: 91 - Distinction
Total Eligible Students: 3

```

Github Repository Link : https://github.com/juug25btech20073-crypto/Python-Programming_25BTRCC031